

Education

Boston University

Kilachand Honors College/College of Engineering

Boston, MA

September 2024 – May 2028

Intended Degree: Mechanical Engineering

Awards/Organizations: F1Tenth ACRO, BU Rocket Propulsion Group, Sikh Association at BU, Theta Tau Professional Engineering Fraternity, QuestBridge Scholar, First Year Innovation Fellowship – Innovate@BU, Society of Asian Scientists and Engineers

Technical Skills

Computer: Python, MATLAB, JavaScript, AutoCAD, SolidWorks, FEA, OnShape, Arduino, Reinforcement Learning, Microsoft Suite

Mechanical: CNC Mill, Drill Press, Laser Cutter, Soldering, 3D Printing, CAM

Languages: English, Punjabi, Hindi, Urdu, Spanish

Projects

BU F1Tenth Autonomous Race Car

December 2024 – Present

Executive Board Member: Hardware

- Leading mechanical design for a Guinness-World Record-attempt autonomous drag car, converting a 2008 Corvette into a fully autonomous platform
- Designed and machined chassis components for 1:10 scale prototype, improving structural rigidity by 30% and reducing weight by 15% through CAD optimization
- Integrated LIDAR, servo control, and sensor systems, achieving 98% hardware-software communication reliability in bench tests
- Modeled and validated vehicle dynamics in CARLA and ROAR simulators, enhancing control precision and path-planning accuracy by 25% in simulation trials

OrthoCAD Pro

July 2025 – September 2025

Personal-Interest Project

- Developed a web-based orthotic design platform integrating computer vision and parametric CAD to automate custom orthotic fabrication
- Implemented a JavaScript-based photogrammetry pipeline achieving a ± 1 mm precision, exceeding clinical tolerance by 50% for smartphone-based 3D foot scans
- Automated CAD generation in Fusion 360 and SolidWorks, producing manufacturing-ready orthotic models with customizable biomechanical parameters

The Deskinator – Autonomous Workspace Robot

September 2025 - Present

Design Lead

- Designed and manufactured an autonomous desktop-cleaning robot using pose-graph estimation and multi-sensor fusion for real-time edge detection and navigation
- Integrated NEMA-17 stepper motors, APDS 9960 proximity sensors, and custom-machined housing components to achieve > 95% edge-tracking accuracy and smooth, controlled motion
- Developed embedded control logic on Raspberry Pi with PID feedback, improving motion stability, odometry, and manufacturability by 40% across prototype iterations

Academic Experience

First-Year Innovation Fellowship

September 2024 – June 2025

2024-2025 Fellow

- Designed prototypes for an insulin delivery system integrating mechanical design with embedded software to model bionic pancreas behavior
- Applied machine learning concepts and control design to biomedical hardware for real-time glucose management

First Robotics (FRC)

September 2020 – May 2024

Pneumatic/Electrical System Co-Lead

- Built a functioning electrical and pneumatic system for an international-level FRC robot. First place in the SBPLI Long-Island Championship, second place at the finger-lakes national invitational event, and 5x FIRST World Competition attendee.

Work Experience

Boston University Chemistry Department: General Chemistry

September 2024 – June 2025

Lab Assistant

- Effectively manage the instructional laboratories in the Chemistry Department by setting up the laboratory assignments before each section
- Deliver supplies, manage chemical inventory, organize and report hazardous waste, and prepare reports regarding laboratory standards.

Leadership

Executive Board Member: BU F1Tenth

December 2024 - Present

Executive Board Member: Sikh Association at Boston University

June 2025 – Present

Academic Chair: Theta Tau Professional Engineering Fraternity

June 2025 – Present